MINI SEARCH ENGINE / INFORMATION-RETRIEVAL SYSTEM

Course: Data Structures and Algorithms – 3

Team Number: 5

Section: 12

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Abstract

- The Mini Search Engine / Information-Retrieval System retrieves relevant information from a collection of documents.
- The system first creates an index of the document collection to make searching more efficient.
- It supports four types of queries:
  - Exact keyword search
  - Phrase/substring search
  - Fuzzy search
  - Document-similarity search
- Retrieved results are ranked according to relevance.
- The project focuses on implementing the core search algorithms manually and analyzing their time and space complexity.

Introduction

- Search engines are used to quickly retrieve useful information from large collections of documents.
- Searching every document repeatedly can become inefficient as the collection grows.
- Our system uses indexing to improve the efficiency of query processing.
- It combines traditional keyword-based searching with similarity-based searching.
- The project demonstrates important DSA concepts such as:
- Indexing
- String matching
- Searching
- Ranking
- Similarity measurement
- The main goal is to balance accuracy, efficiency, scalability, and complexity.

Software & Hardware Requirements

Software Requirements

- Programming Language: Java
- Operating System: macOS / Windows / Linux
- IDE: IntelliJ IDEA / VS Code / Eclipse
- JDK: Java Development Kit
- Input: Text document collection
- Data Structures: HashMap, ArrayList, and suitable indexing structures

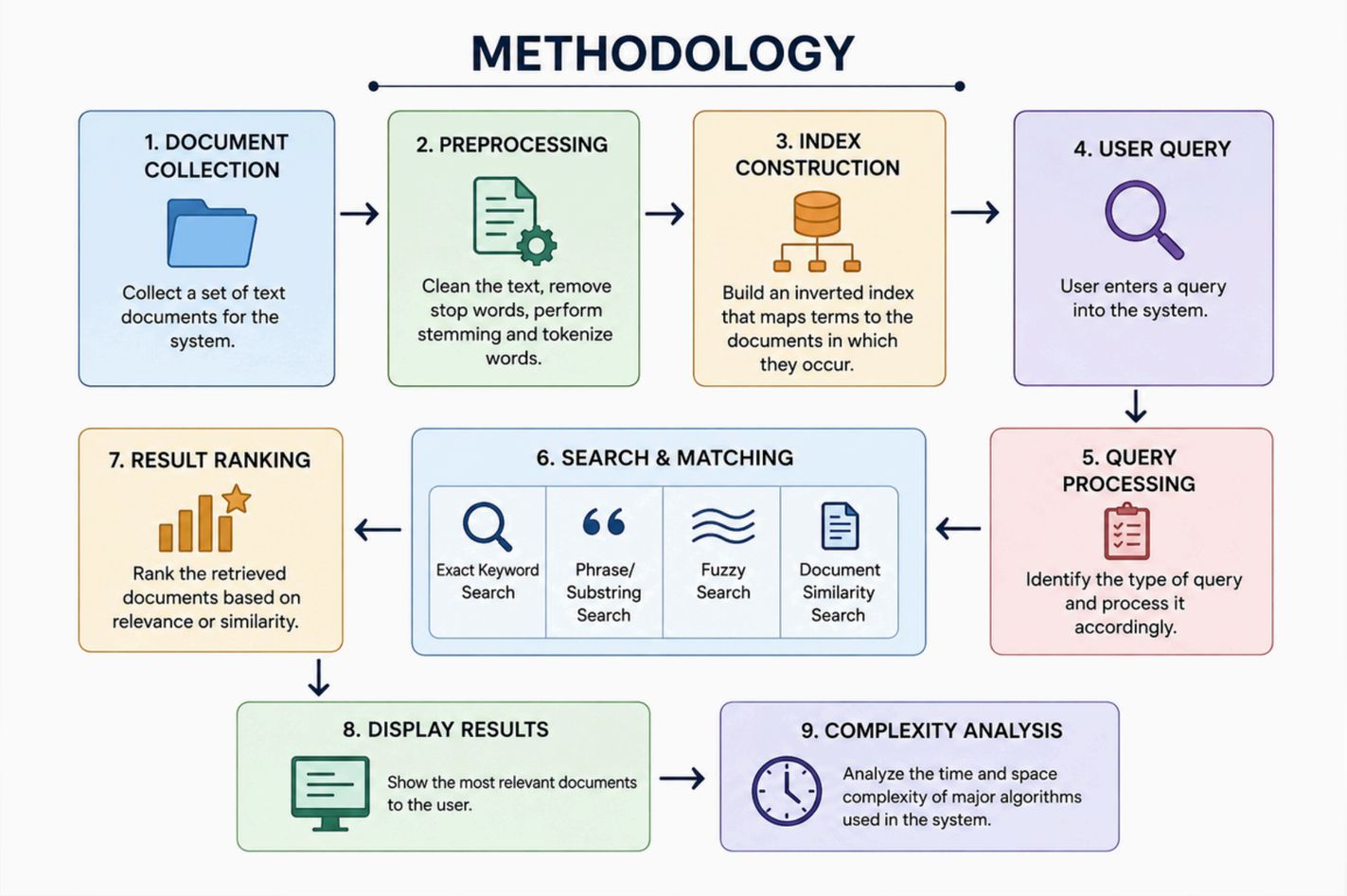
Hardware Requirements

- Processor: Intel/AMD or Apple Silicon
- RAM: Minimum 4 GB
- Storage: Minimum 500 MB
- Display: Laptop/Monitor
- Keyboard & Mouse/Trackpad

Literature Survey

Technique	Purpose
Keyword Search	Finds documents containing specific terms
Inverted Indexing	Quickly maps terms to relevant documents
String Matching	Finds phrases or substrings
Fuzzy Search	Handles spelling mistakes and approximate matches
Similarity Search	Finds documents similar to a query/document
Ranking	Places the most relevant results first

Methodology



Conclusion

- The project demonstrates the basic working of a Mini Search Engine using Data Structures and Algorithms.
- An inverted index improves the efficiency of document retrieval.
- The system supports exact, phrase/substring, fuzzy, and document-similarity searches.
- Results are ranked based on relevance or similarity.
- The project provides practical understanding of searching, indexing, string matching, ranking, and similarity measurement.
- The system also evaluates the time and space complexity of the implemented algorithms.

THANK YOU